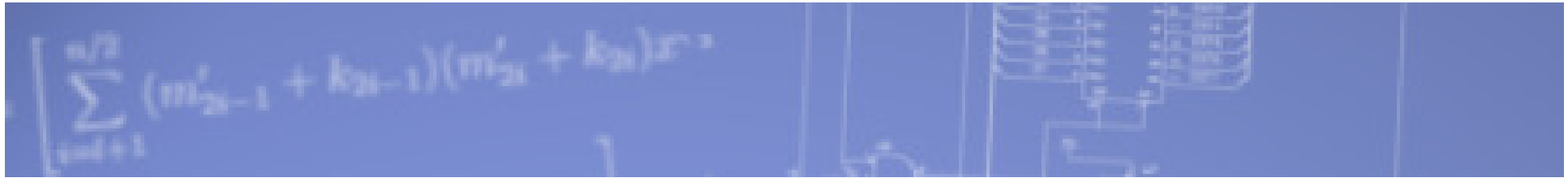


0113611 Computer Hardware

Digital Logic Review

Textbook References

- Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with VHDL Design*, 3rd Edition
- M. M. Mano and C. R. Kime. (2008) Logic and Computer Design Fundamentals, 4th Edition. Prentice Hall. (ISBN: 0-13-600158-0).
- OR your undergraduate digital logic textbook
- Adapted from lecture notes at ece.gmu.edu/.../ECE/ECE545

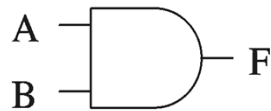


Basic Logic Review

Basic Concepts

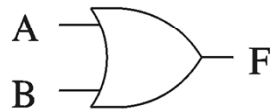
- Simple logic gates
 - AND $\rightarrow$ 0 if one or more inputs is 0
 - OR $\rightarrow$ 1 if one or more inputs is 1
 - NOT
 - NAND = AND + NOT
 - 1 if one or more inputs is 0
 - NOR = OR + NOT
 - 0 if one or more input is 1
 - XOR implements exclusive-OR function
- NAND and NOR gates require fewer transistors than AND and OR in standard CMOS
- Functionality can be expressed by a truth table
 - A truth table lists output for each possible input combination

Basic Logic Gates



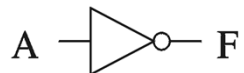
AND gate

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1



OR gate

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

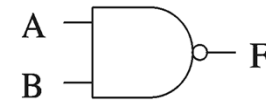


NOT gate

A	F
0	1
1	0

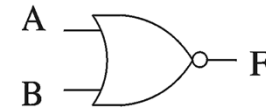
Logic symbol

Truth table



NAND gate

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0



NOR gate

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0



XOR gate

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

Logic symbol

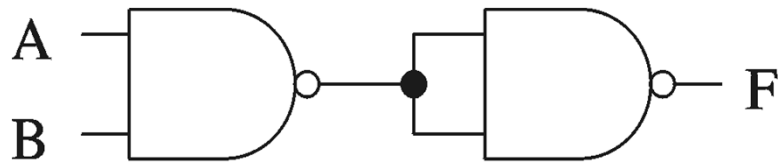
Truth table

Complete Set of Gates

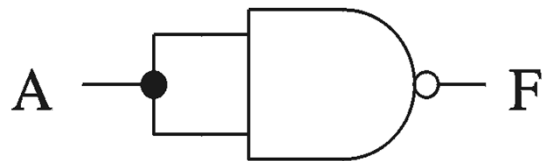
- Complete sets
 - A set of gates is complete
 - if we can implement any logical function using only the type of gates in the set
 - Some example complete sets
 - {AND, OR, NOT} ← Not a minimal complete set
 - {AND, NOT}
 - {OR, NOT}
 - {NAND}
 - {NOR}
 - Minimal complete set
 - A complete set with no redundant elements.

NAND as a Complete Set

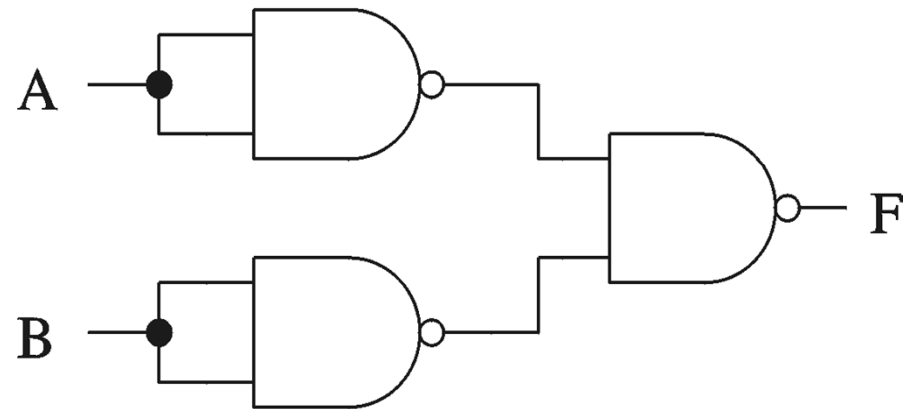
- Proving NAND gate is universal



AND gate



NOT gate



OR gate

Logic Functions

- Logical functions can be expressed in several ways:
 - Truth table
 - Logical expressions
 - Graphical form
 - HDL code
- Example:
 - Majority function
 - Output is one whenever majority of inputs is 1
 - We use 3-input majority function

Logic Functions (cont'd)

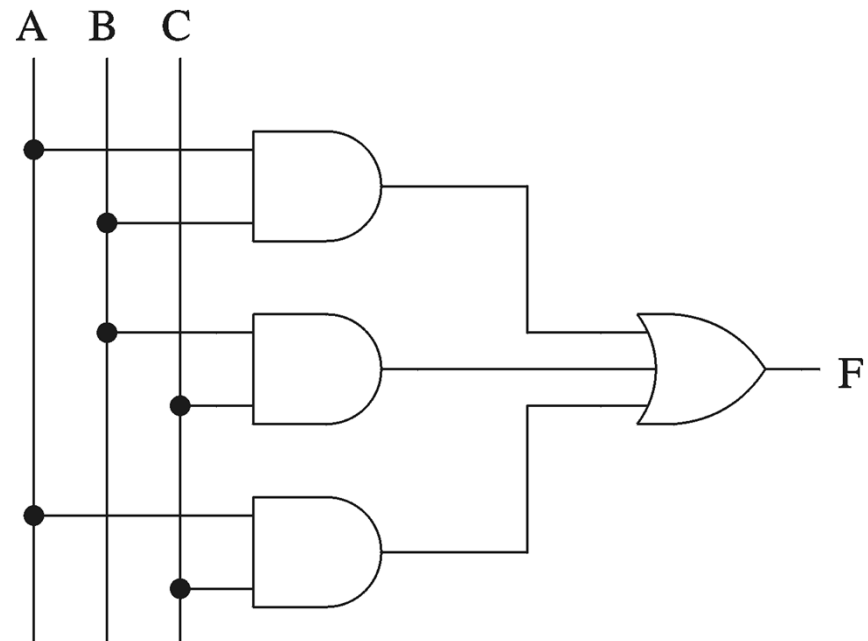
Truth table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Logical expression form

$$F = A B + B C + A C$$

Graphical schematic form



Boolean Algebra

Boolean identities

Name	AND version	OR version
Identity	$x \cdot 1 = x$	$x + 0 = x$
Complement	$x \cdot x' = 0$	$x + x' = 1$
Commutative	$x \cdot y = y \cdot x$	$x + y = y + x$
Distribution	$x \cdot (y + z) = xy + xz$	$x + (y \cdot z) =$ $(x + y) (x + z)$
Idempotent	$x \cdot x = x$	$x + x = x$
Null	$x \cdot 0 = 0$	$x + 1 = 1$

Boolean Algebra (cont'd)

- Boolean identities (cont'd)

Name	AND version	OR version
Involution	$x = (x')'$	---
Absorption	$x \cdot (x + y) = x$	$x + (x \cdot y) = x$
Associative	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	$x + (y + z) =$ $(x + y) + z$
de Morgan	$(x \cdot y)' = x' + y'$	$(x + y)' = x' \cdot y'$

(de Morgan's law in particular is very useful)

Majority Function Using Other Gates

- Using NAND gates
 - Get an equivalent expression

$$A B + C D = (A B + C D)''$$

- Using de Morgan's law

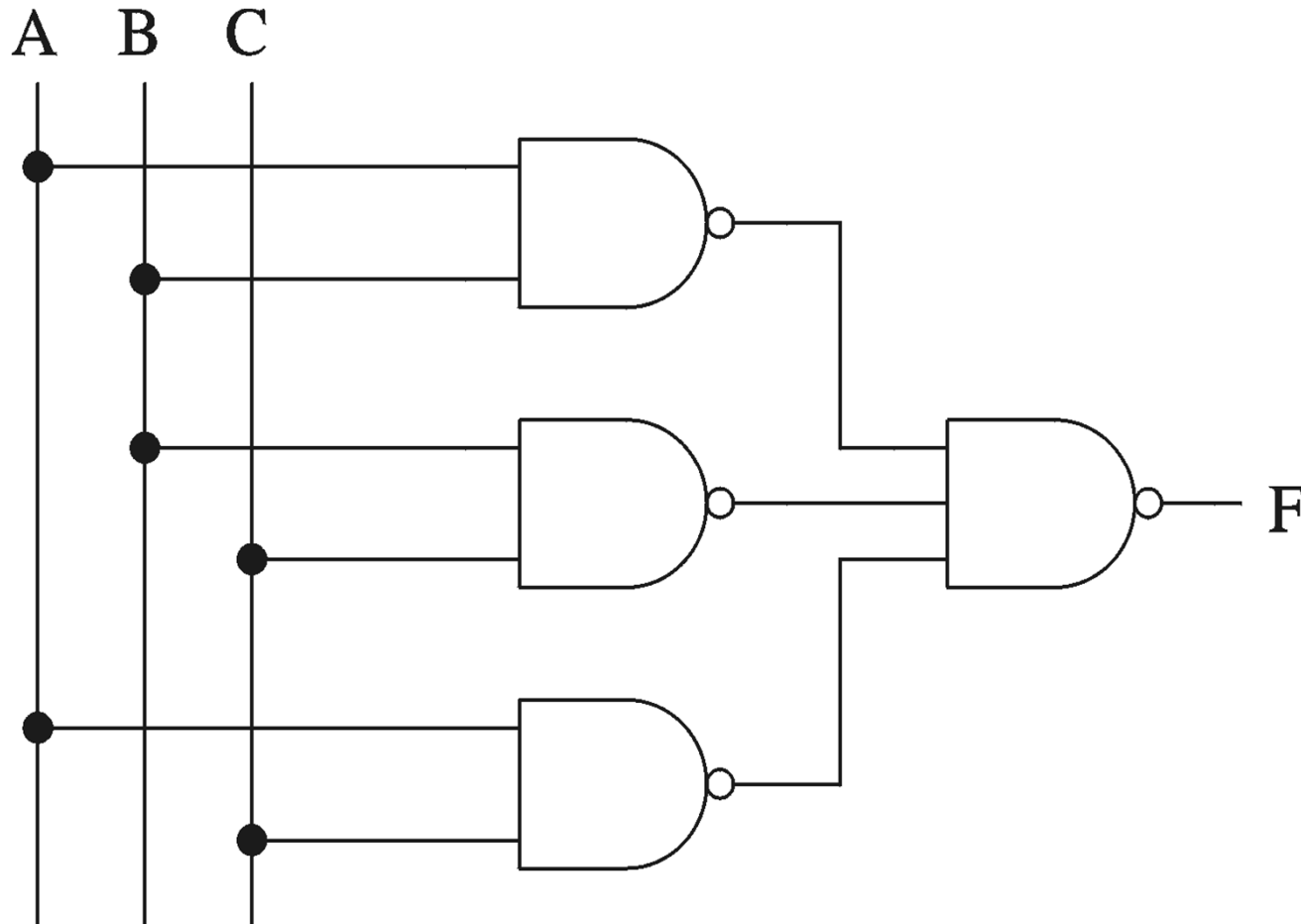
$$A B + C D = ((A B)' \cdot (C D)')'$$

- Can be generalized
 - Example: Majority function

$$A B + B C + A C = ((A B)' \cdot (B C)' \cdot (A C)')'$$

Majority Function Using Other Gates (cont'd)

- Majority function



Karnaugh Maps

x_1	x_2	x_3	
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

(a) Truth table

		$x_1 x_2$			
		00	01	11	10
x_3	0	m_0	m_2	m_6	m_4
	1	m_1	m_3	m_7	m_5

(b) Karnaugh map

		$x_1 x_2$			
		00	01	11	10
x_3	0	1	1	1	1
	1	0	0	0	1

$f = \bar{x}_3 + x_1 \bar{x}_2$

An example of three-variable Karnaugh maps

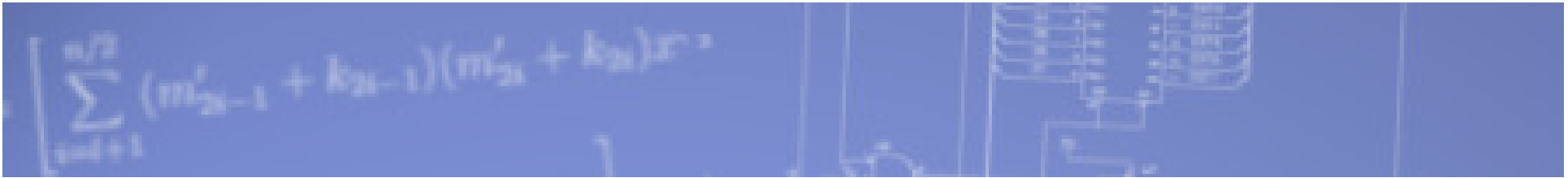
Numbers

Decimal	Binary	Octal	Hexadecimal
00	00000	00	00
01	00001	01	01
02	00010	02	02
03	00011	03	03
04	00100	04	04
05	00101	05	05
06	00110	06	06
07	00111	07	07
08	01000	10	08
09	01001	11	09
10	01010	12	0A
11	01011	13	0B
12	01100	14	0C
13	01101	15	0D
14	01110	16	0E
15	01111	17	0F
16	10000	20	10
17	10001	21	11
18	10010	22	12

Table 5.1 Interpretation of four-bit signed integers.

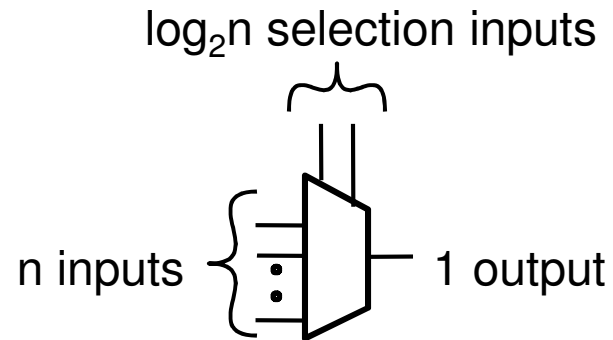
$b_3b_2b_1b_0$	Sign and magnitude	1's complement	2's complement
0111	+7	+7	+7
0110	+6	+6	+6
0101	+5	+5	+5
0100	+4	+4	+4
0011	+3	+3	+3
0010	+2	+2	+2
0001	+1	+1	+1
0000	+0	+0	+0
1000	-0	-7	-8
1001	-1	-6	-7
1010	-2	-5	-6
1011	-3	-4	-5
1100	-4	-3	-4
1101	-5	-2	-3
1110	-6	-1	-2
1111	-7	-0	-1

Numbers in different systems



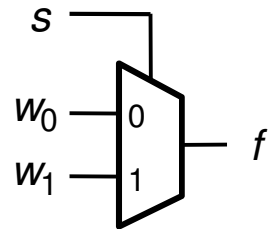
Combinational Logic Building Blocks

Multiplexers



- multiplexer
 - n binary inputs (binary input = 1-bit input)
 - $\log_2 n$ binary selection inputs
 - 1 binary output
 - Function: one of n inputs is placed onto output
 - Called **n-to-1** multiplexer

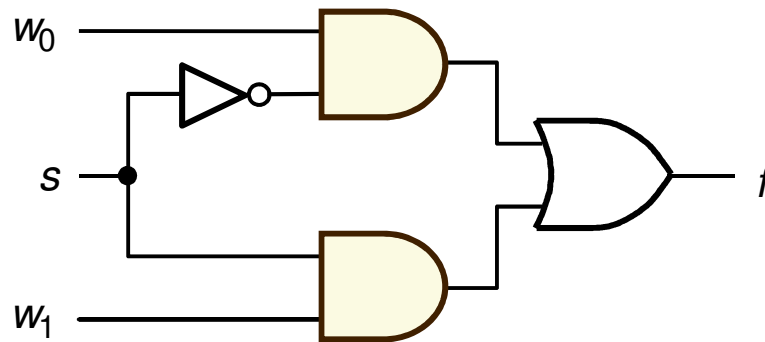
2-to-1 Multiplexer



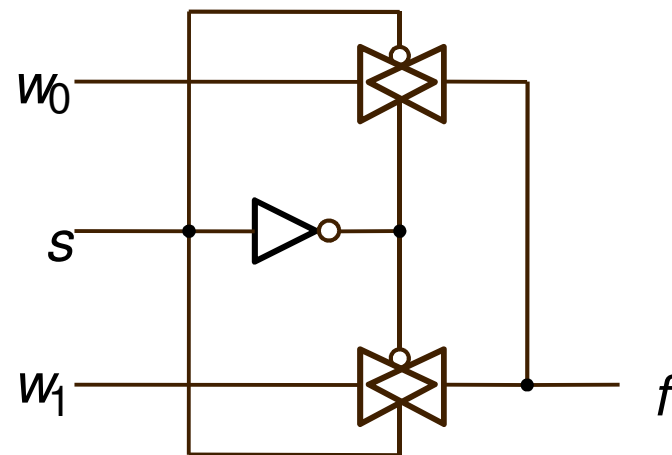
(a) Graphical symbol

s	f
0	w_0
1	w_1

(b) Truth table

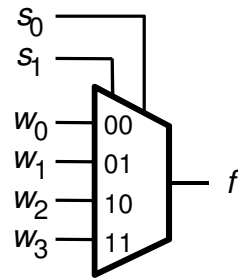


(c) Sum-of-products circuit



(d) Circuit with transmission gates

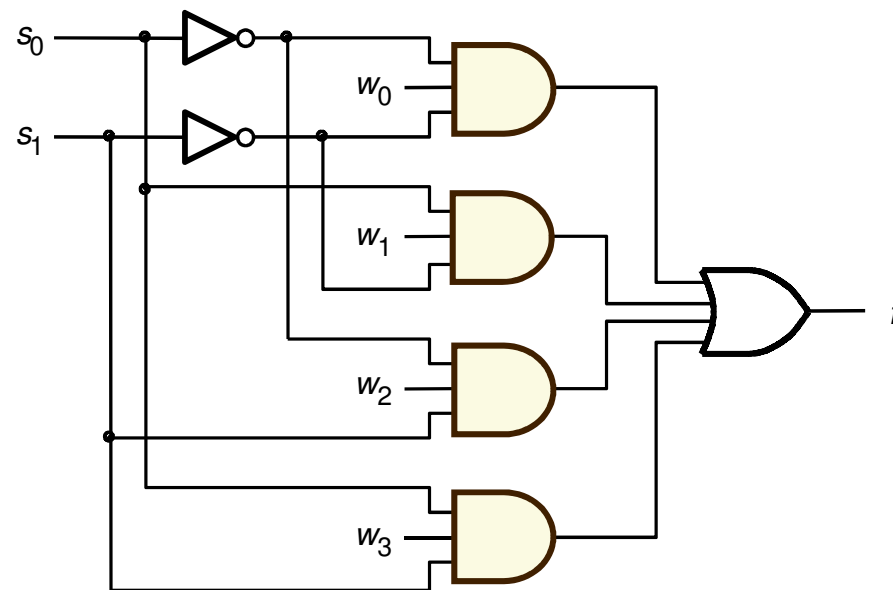
4-to-1 Multiplexer



(a) Graphic symbol

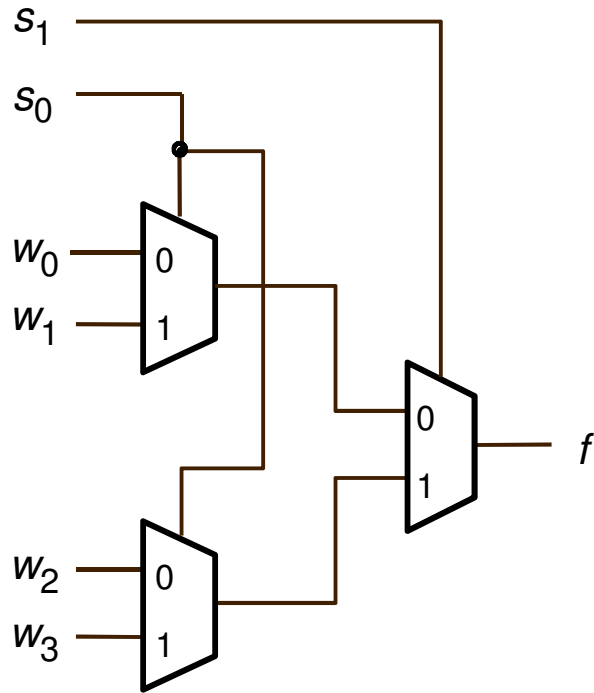
s_1	s_0	f
0	0	w_0
0	1	w_1
1	0	w_2
1	1	w_3

(b) Truth table

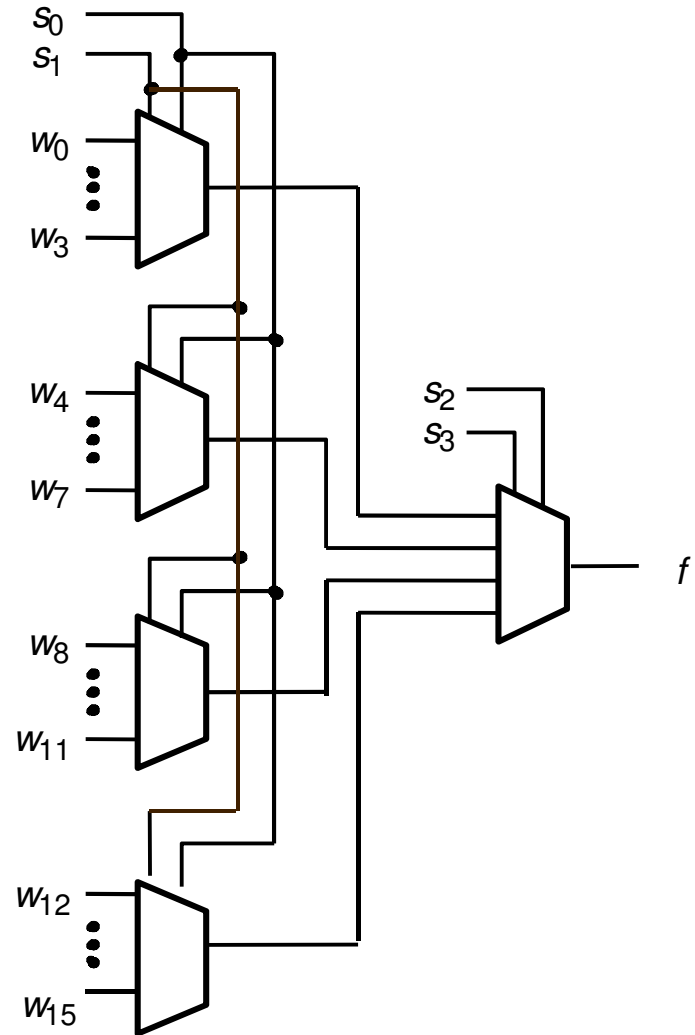


(c) Circuit

Multiplexer

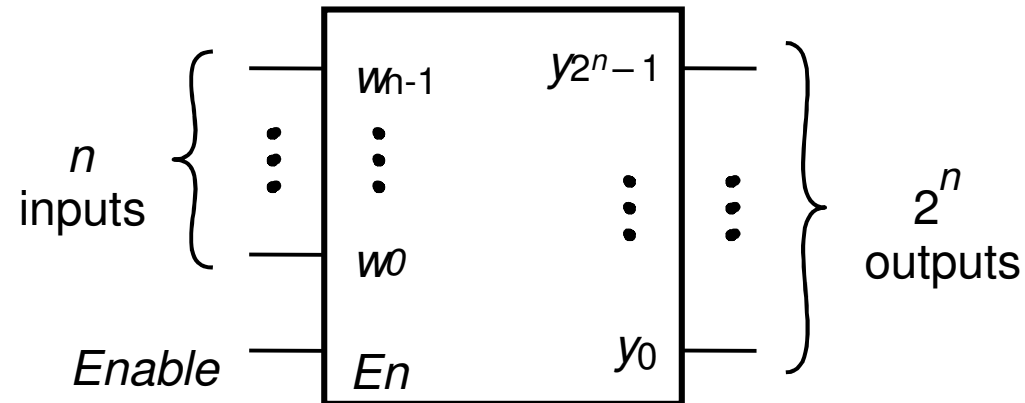


Using 2-to-1 multiplexers to build a 4-to-1 multiplexer.



A 16-to-1 multiplexer.

Decoders

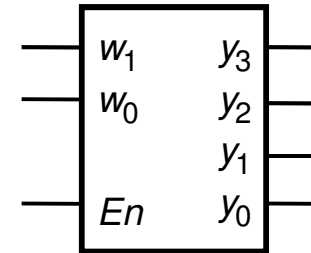


- **Decoder**
 - n binary inputs
 - 2^n binary outputs
 - Function: decode encoded information
 - If enable=1, one output is asserted high, the other outputs are asserted low
 - If enable=0, all outputs asserted low
 - Often, enable pin is not needed (i.e. the decoder is always enabled)
 - Called **n-to- 2^n** decoder
 - Can consider n binary inputs as a single n -bit input
 - Can consider 2^n binary outputs as a single 2^n -bit output
 - Decoders are often used for RAM/ROM addressing

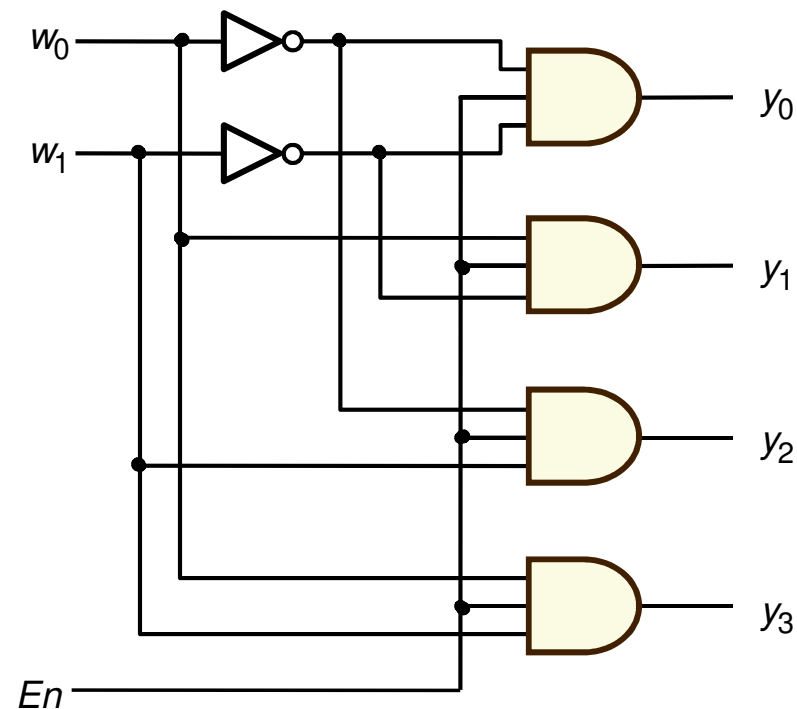
2-to-4 Decoder

En	w_1	w_0	y_3	y_2	y_1	y_0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0
0	-	-	0	0	0	0

(a) Truth table

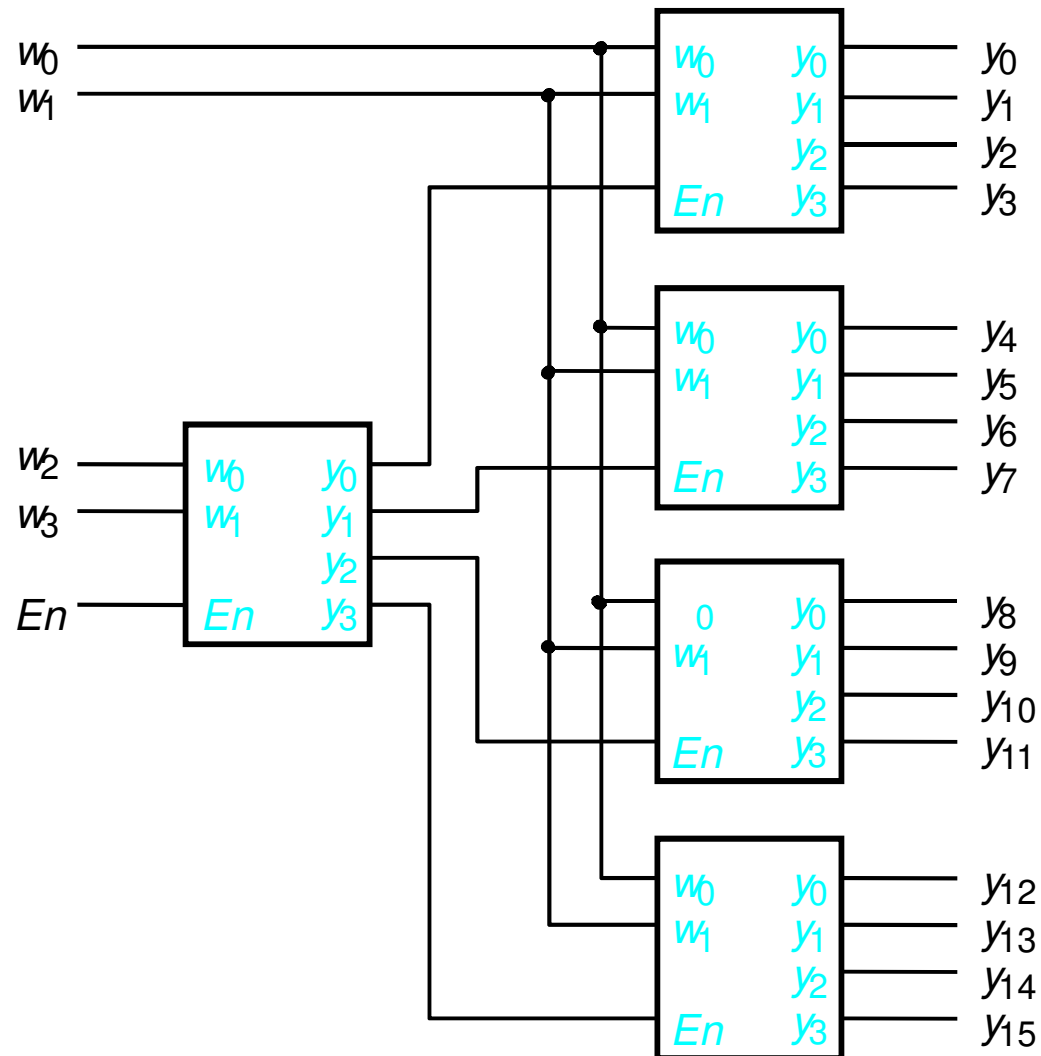


(b) Graphical symbol



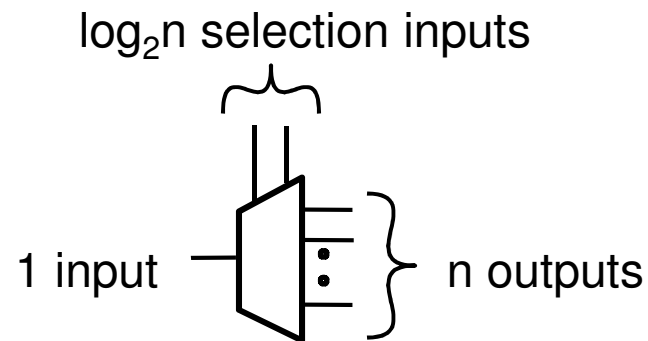
(c) Logic circuit

Decoders



A 4-to-16 decoder built using a decoder tree

Demultiplexers

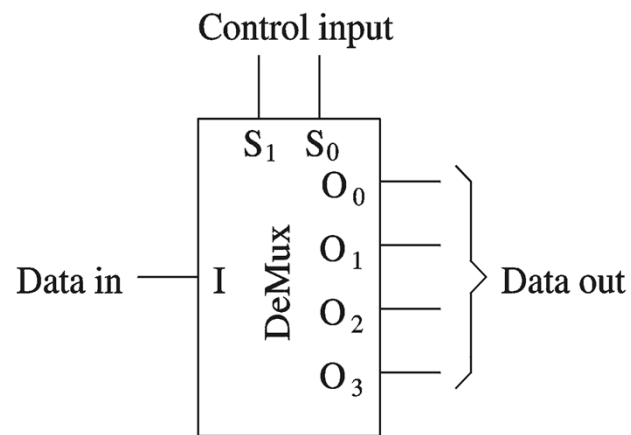


- Demultiplexer
 - 1 binary input
 - n binary outputs
 - $\log_2 n$ binary selection inputs
 - Function: places input onto one of n outputs, with the remaining outputs asserted low
 - Called **1-to- n** demultiplexer
- Closely related to decoder
 - Can build 1-to- n demultiplexer from $\log_2 n$ -to- n decoder by using the decoder's enable signal as the demultiplexer's input signal, and using decoder's input signals as the demultiplexer's selection input signals.

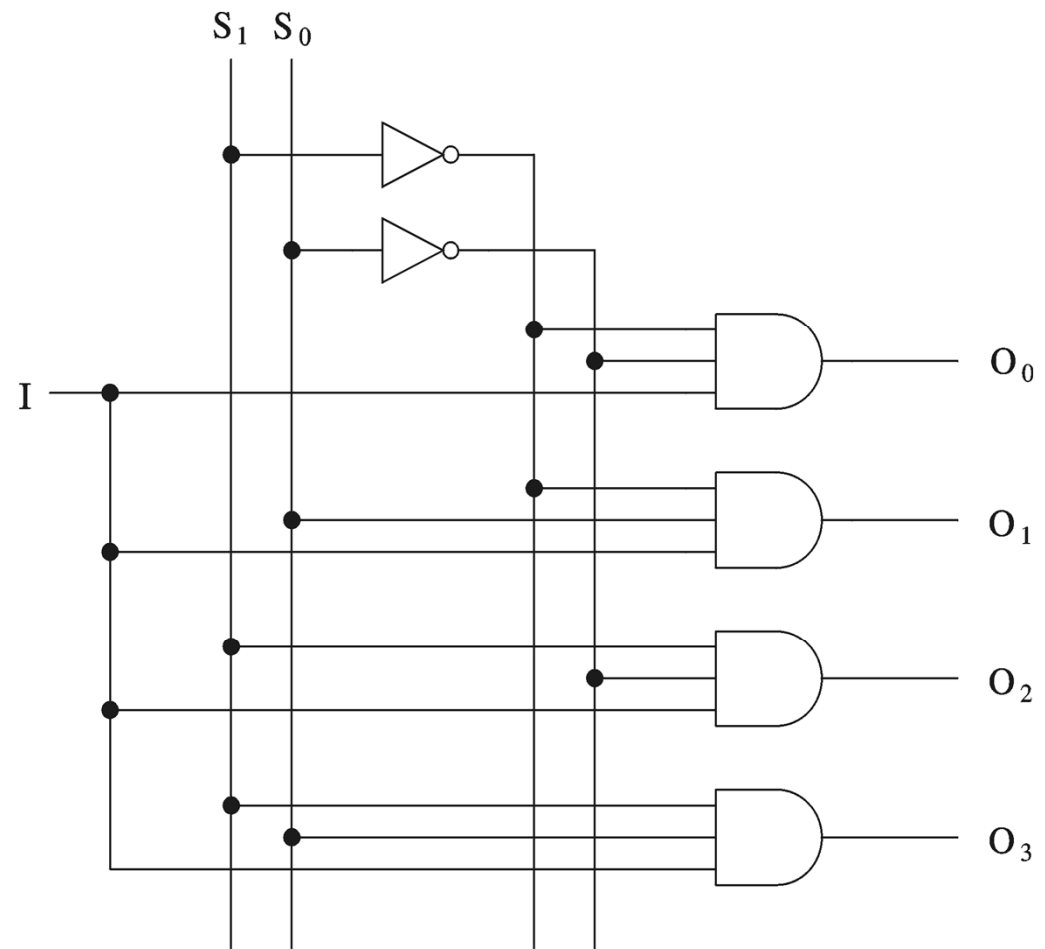
1-to-4 Demultiplexer

$S_1 S_0$		$Q_3 Q_2 Q_1 Q_0$			
1	0	3	2	1	0
0	0	0	0	0	A
0	1	0	0	A	0
1	0	0	A	0	0
1	1	A	0	0	0
-	-	0	0	0	0

(a) Truth table

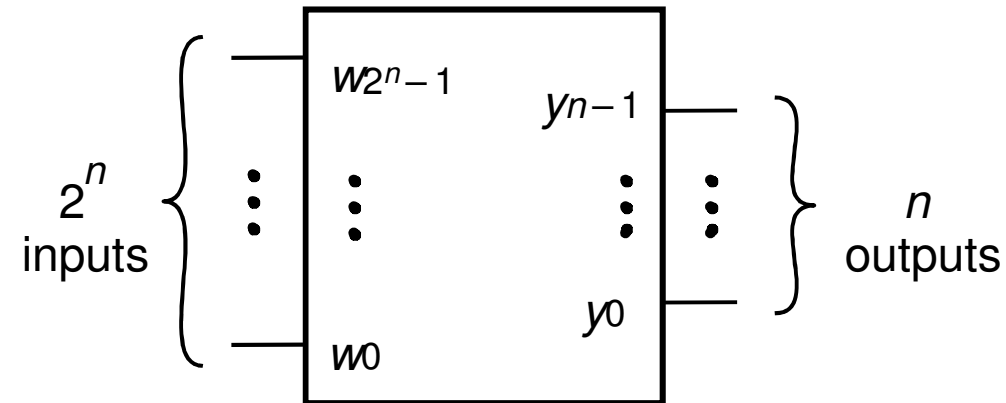


(b) Graphic symbol



(c) Circuit

Encoders

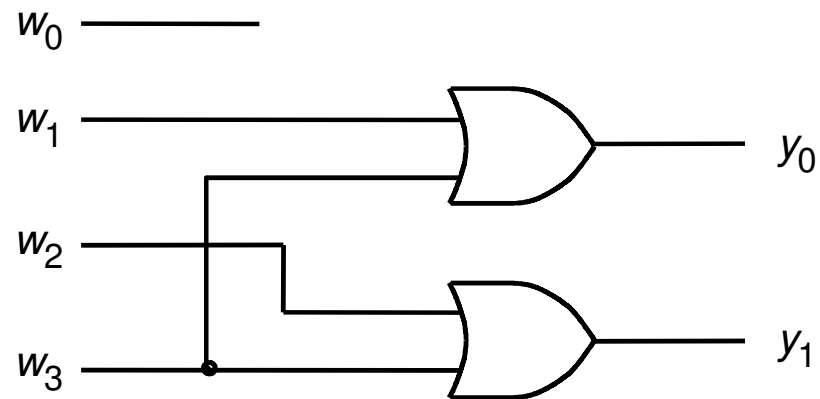


- Encoder
 - 2ⁿ binary inputs
 - n binary outputs
 - Function: encodes information into an n -bit code
 - Called **2ⁿ-to- n** encoder
 - Can consider 2ⁿ binary inputs as a single 2ⁿ-bit input
 - Can consider n binary output as a single n -bit output
- Encoders only work when **exactly one** binary input is equal to 1

4-to-2 Encoder

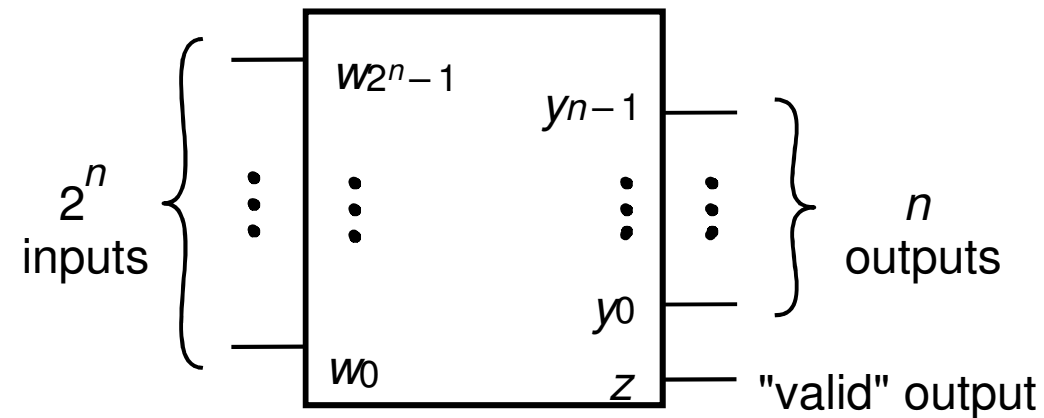
w_3	w_2	w_1	w_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

(a) Truth table



(b) Circuit

Priority Encoders



- Priority Encoder
 - 2^n binary inputs
 - n binary outputs
 - 1 binary "valid" output
 - Function: encodes information into an n -bit code based on priority of inputs
 - Called **2^n -to- n** priority encoder
- Priority encoder allows for multiple inputs to have a value of '1', as it encodes the input with the highest priority (MSB = highest priority, LSB = lowest priority)
 - "valid" output indicates when priority encoder output is valid
 - Priority encoder is more common than an encoder

4-to-2 Priority Encoder

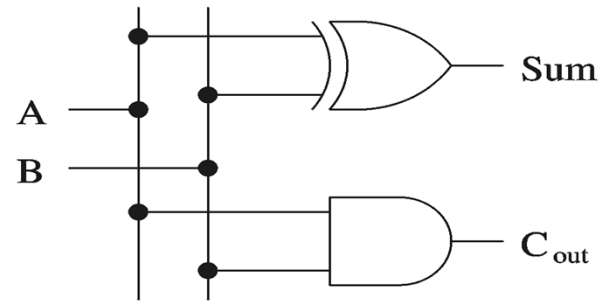
w_3	w_2	w_1	w_0	y_1	y_0	z
0	0	0	0	-	-	0
0	0	0	1	0	0	1
0	0	1	-	0	1	1
0	1	-	-	1	0	1
1	-	-	-	1	1	1

Single-Bit Adders

- Half-adder
 - Adds two binary (i.e. 1-bit) inputs A and B
 - Produces a *sum* and *carryout*
 - Problem: Cannot use it alone to build larger adders
- Full-adder
 - Adds three binary (i.e. 1-bit) inputs A , B , and *carryin*
 - Like half-adder, produces a *sum* and *carryout*
 - Allows building M -bit adders ($M > 1$)
 - Simple technique
 - Connect C_{out} of one adder to C_{in} of the next
 - These are called *ripple-carry adders*
 - Shown in next section

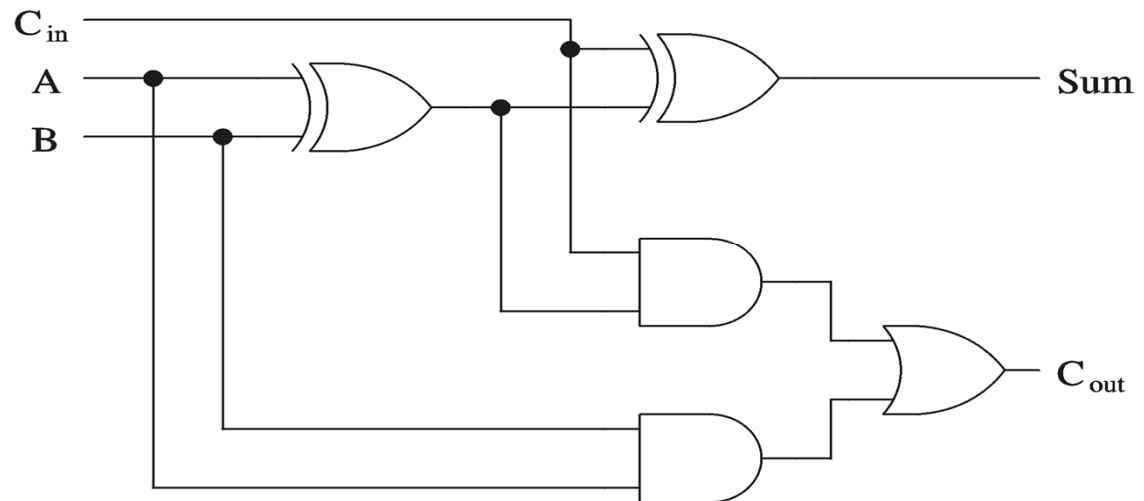
Single-Bit Adders (cont'd)

A	B	Sum	C _{out}
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



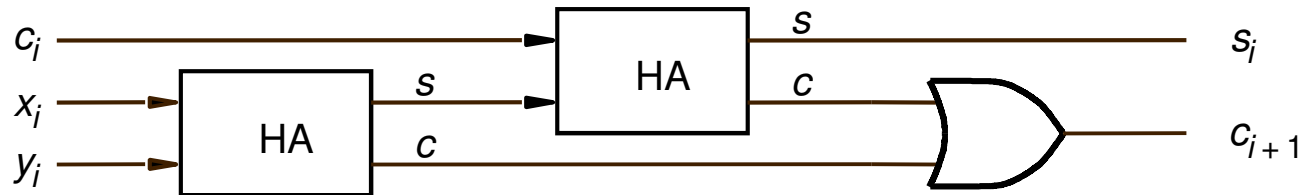
(a) Half-adder truth table and implementation

A	B	C _{in}	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

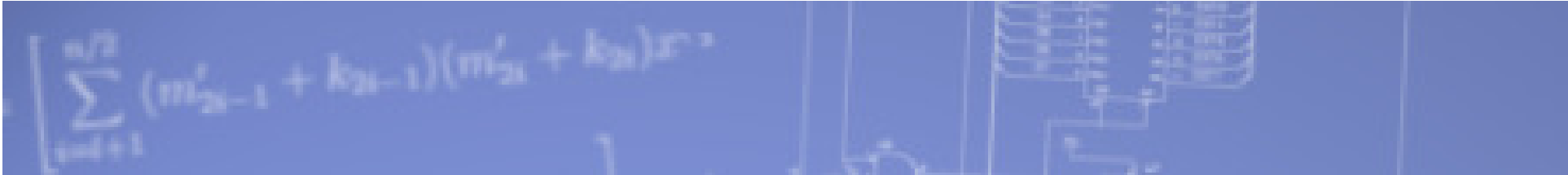


(b) Full-adder truth table and implementation

Adders

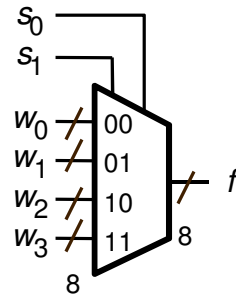


A decomposed implementation of the full-adder circuit



Multi-Bit Combinational Logic Building Blocks

Multi-bit 4-to-1 Multiplexer



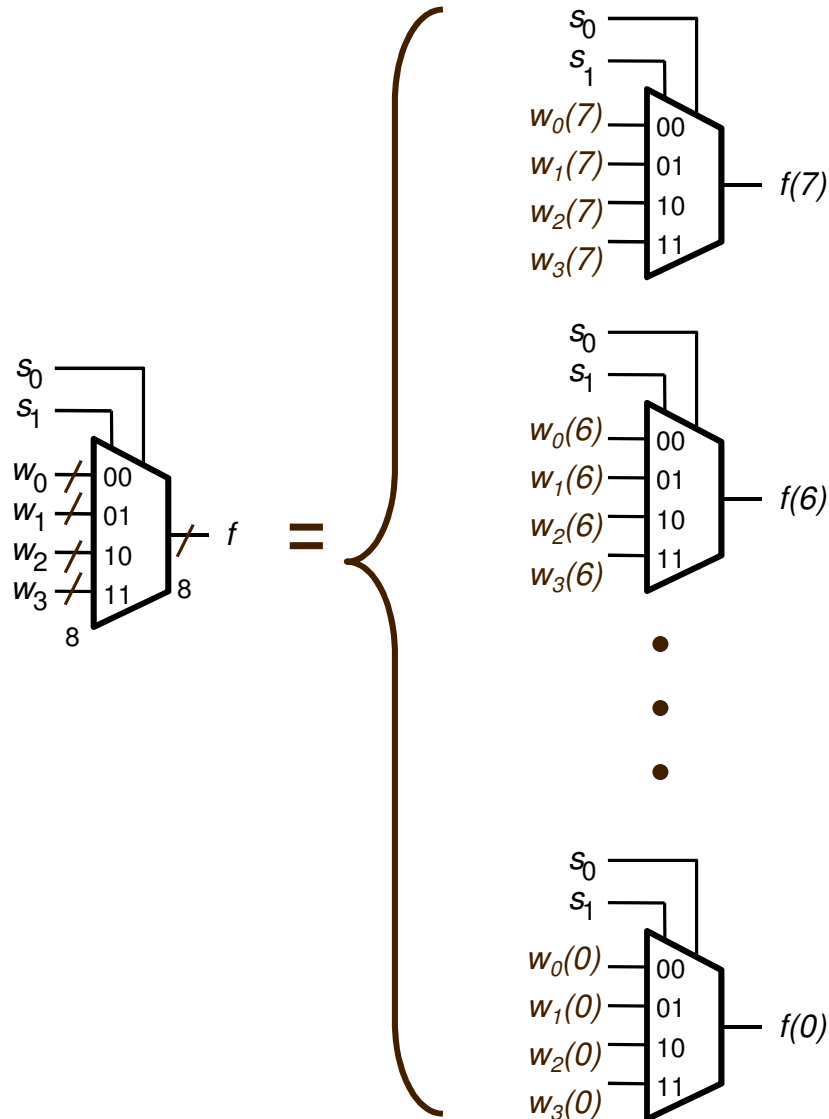
(a) Graphic symbol

s_1	s_0	f
0	0	w_0
0	1	w_1
1	0	w_2
1	1	w_3

(b) Truth table

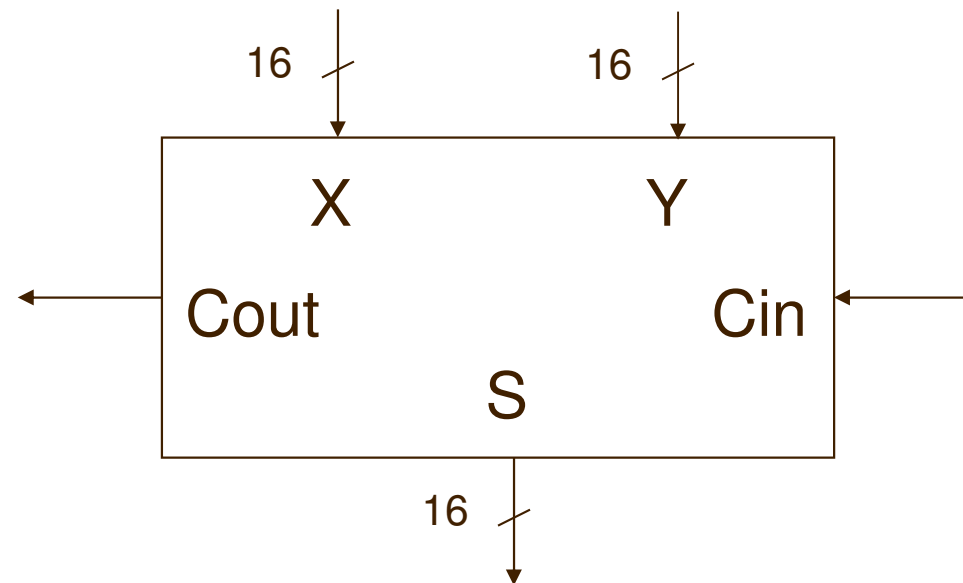
- When drawing schematics, can draw **multi-bit** multiplexers
- Example: 4-to-1 (8 bit) multiplexer
 - 4 inputs (each 8 bits)
 - 1 output (8 bits)
 - 2 selection bits
- Can also have multi-bit 2-to-1 muxes, 16-to-1 muxes, etc.

4-to-1 (8-bit) Multiplexer



A 4-to-1 (8-bit) multiplexer is composed of eight 4-to-1 (1-bit) multiplexers

16-bit Unsigned Adder

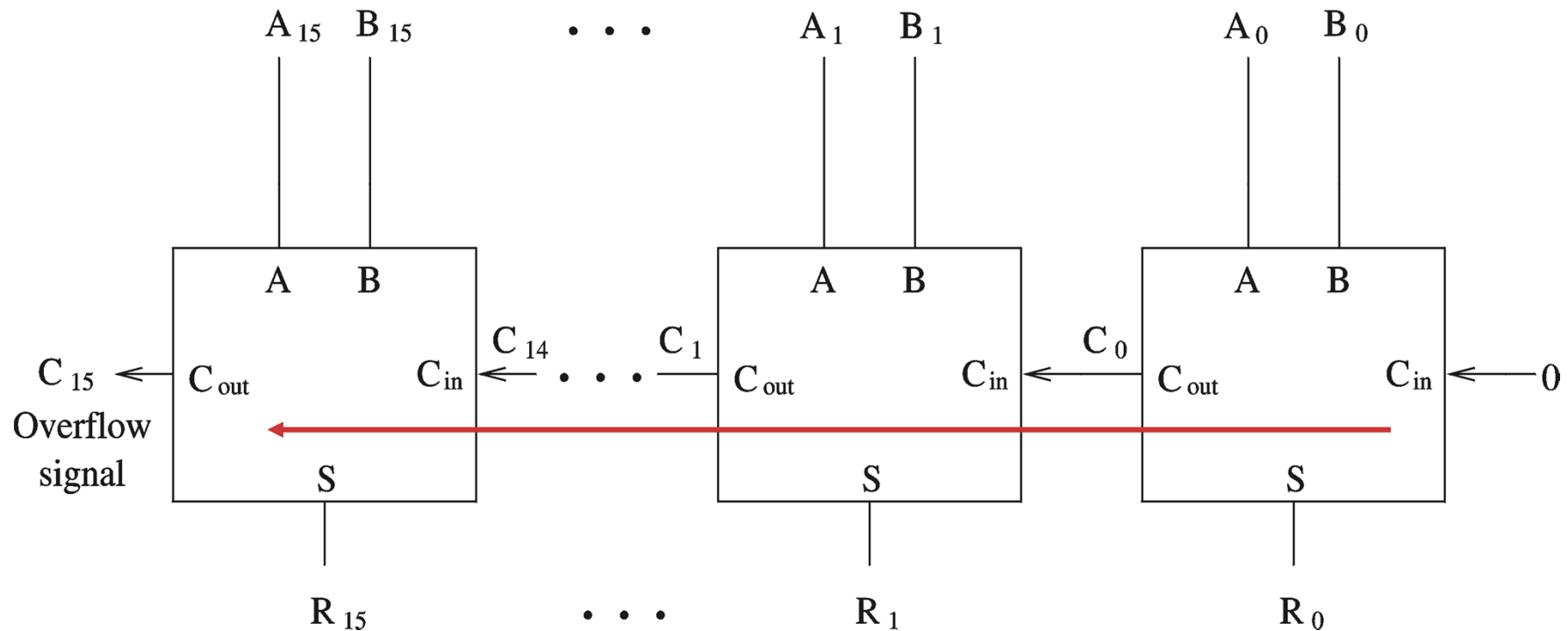


Multi-Bit Ripple-Carry Adder

A 16-bit ripple-carry adder is composed of 16 (1-bit) full adders

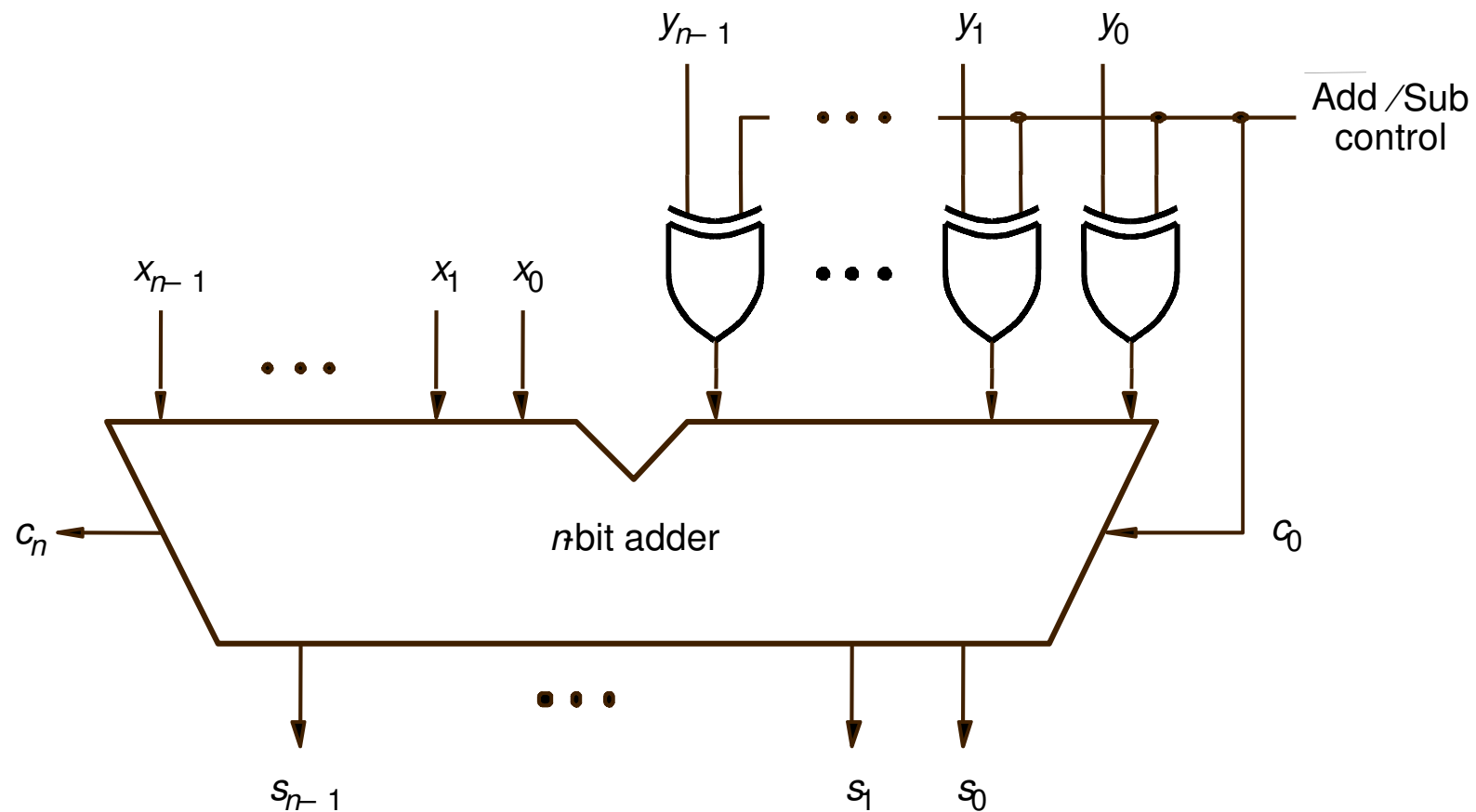
Inputs: 16-bit A, 16-bit B, 1-bit carryin (set to zero in the figure below)

Outputs: 16-bit sum R, 1-bit overflow



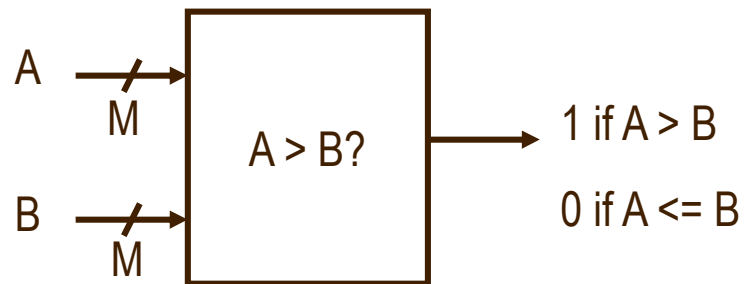
Called a ripple-carry adder because carry ripples from one full-adder to the next.
Critical path is 16 full-adders.

Adder/subtractor unit

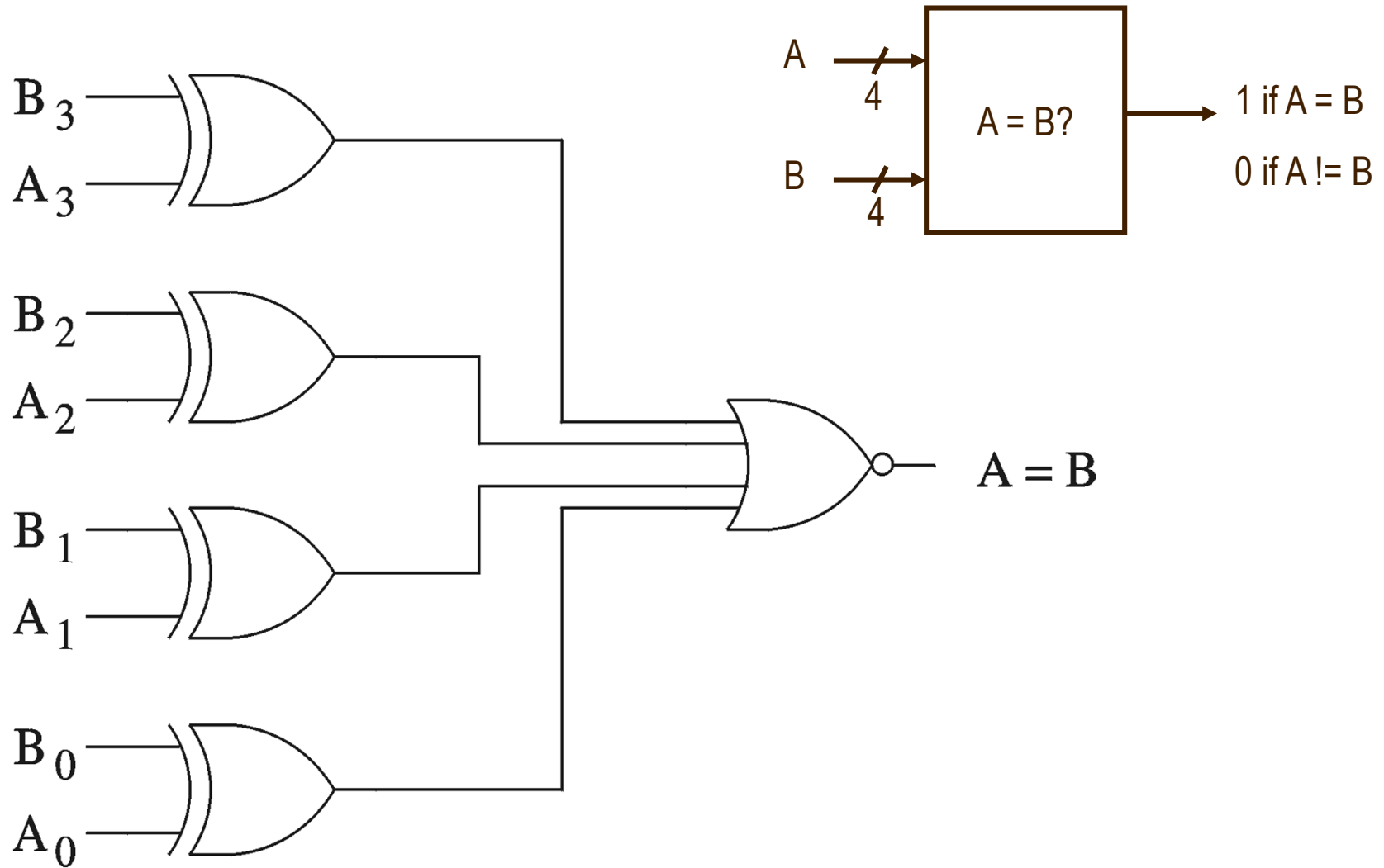


Comparator

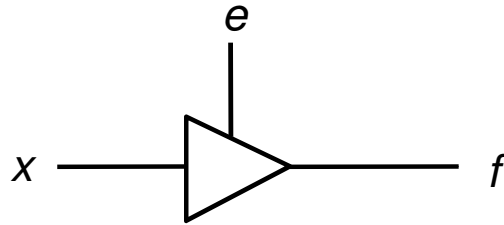
- Used to compare two M-bit numbers and produce a flag ($M > 1$)
 - Inputs: M-bit input A, M-bit input B
 - Output: 1-bit output flag
 - 1 indicates condition is met
 - 0 indicates condition is not met
 - Can compare: $>$, $\geq$, $<$, $\leq$, $=$, etc.



Example: 4-bit comparator ($A = B$)



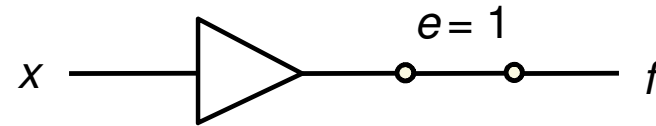
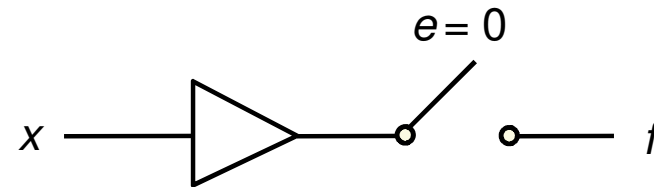
Tri-state Buffer



(a) A tri-state buffer

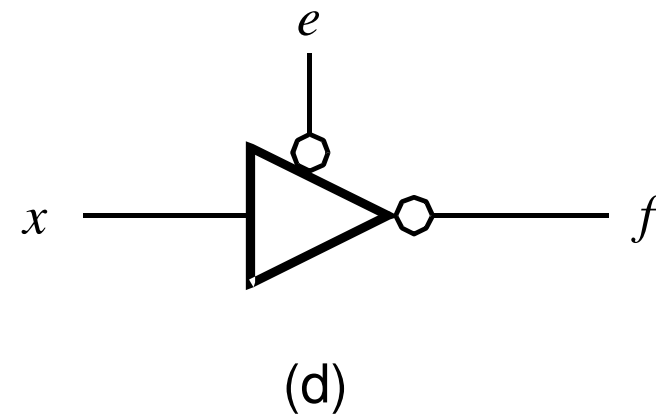
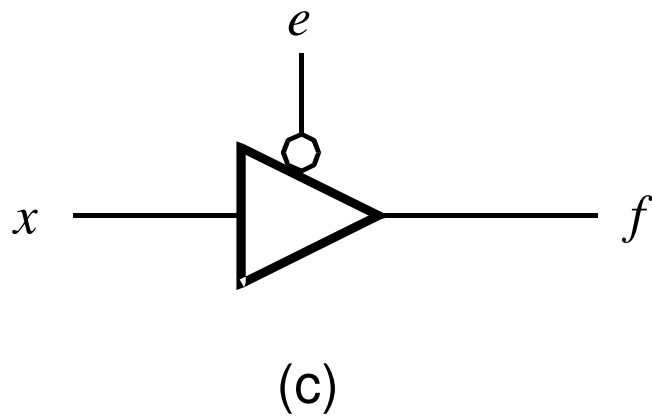
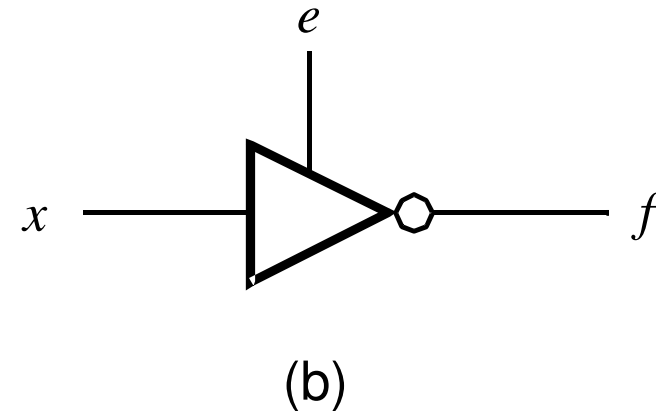
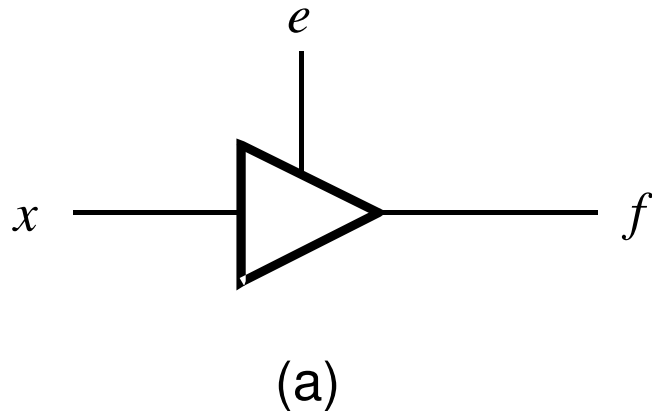
e	x	f
0	0	Z
0	1	Z
1	0	0
1	1	1

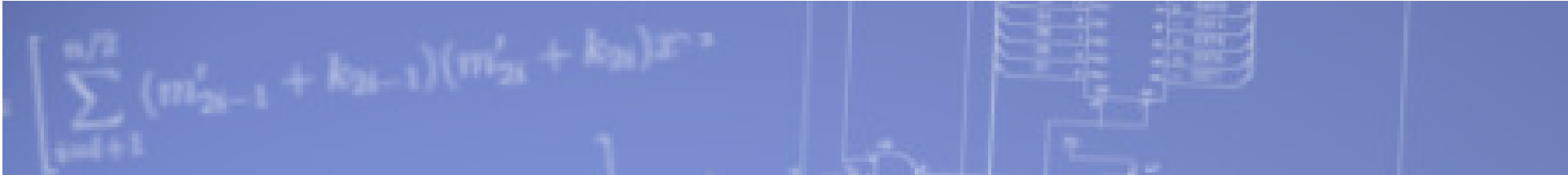
(c) Truth table



(b) Equivalent circuit

Four types of Tri-state Buffers





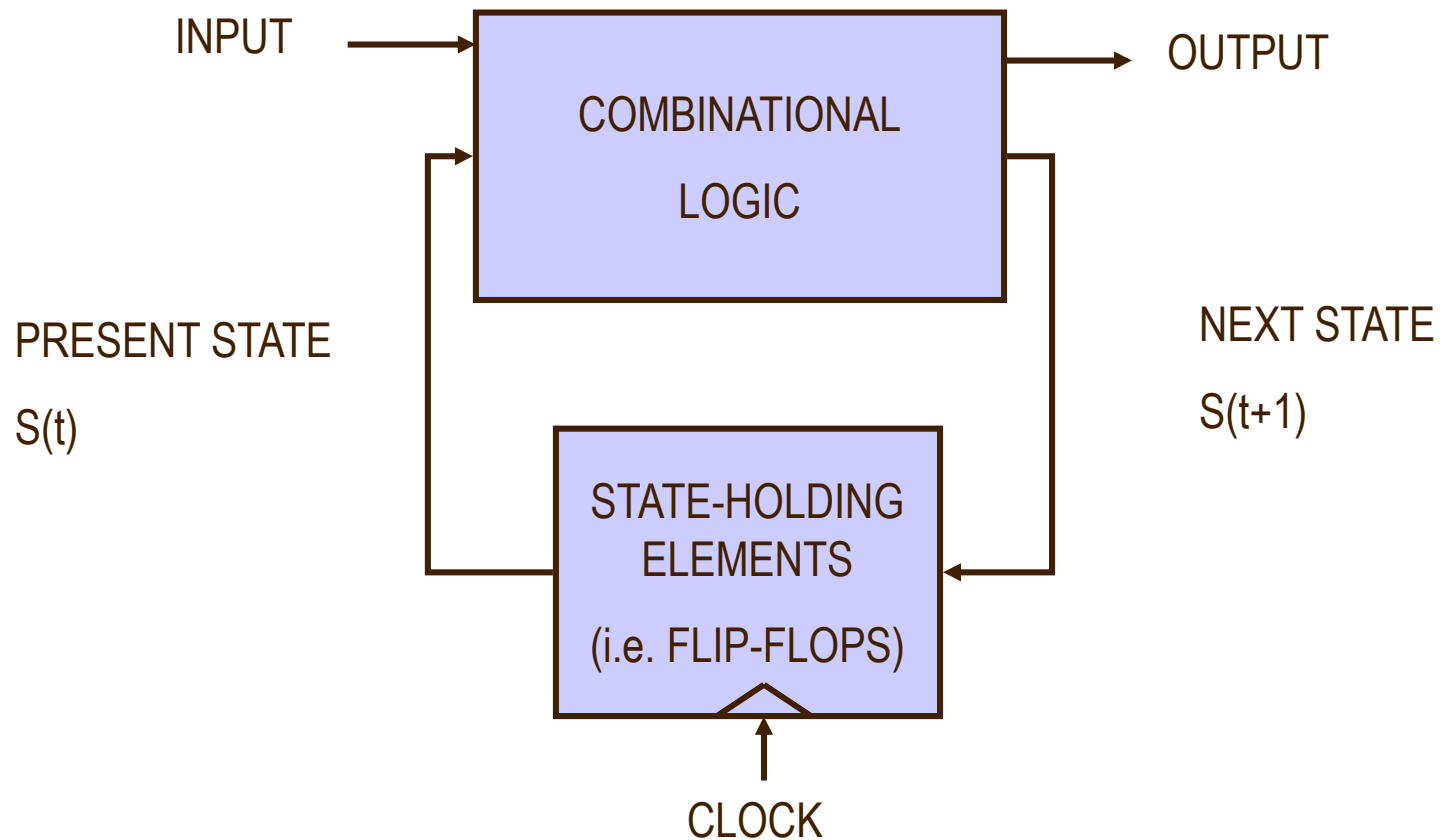
Sequential Logic Building Blocks

Introduction to Sequential Logic

- Output depends on current as well as past inputs
 - Depends on the history
 - Have “memory” property
- Sequential circuit consists of
 - Combinational circuit
 - Feedback circuit
- Past input is encoded into a set of state variables
 - Uses feedback (to feed the state variables)
 - Simple feedback
 - Uses flip flops

Introduction (cont'd)

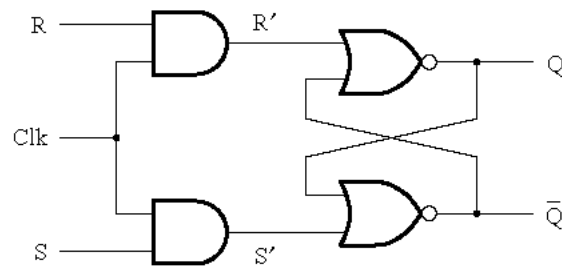
Main components of a typical synchronous sequential circuit
(synchronous = uses a clock to keep circuits in lock step)



State-Holding Memory Elements

- Latch versus Flip Flop
 - Latches are level-sensitive: whenever clock is high, latch is transparent
 - Flip-flops are edge-sensitive: data passes through (i.e. data is sampled) only on a rising (or falling) edge of the clock
 - Latches cheaper to implement than flip-flops
 - Flip-flops are easier to design with than latches
- In this course, primarily use D flip-flops

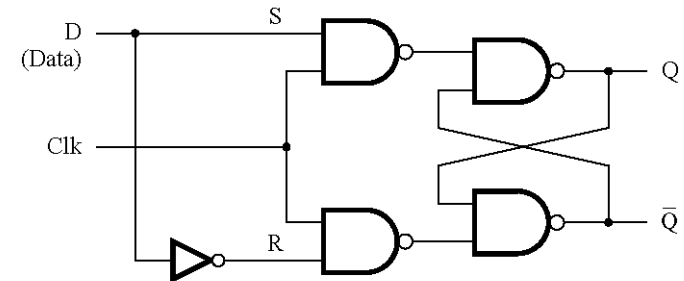
Latches and Flip-Flops



(a) Circuit

Clk	S	R	$Q(t+1)$
0	x	x	$Q(t)$ (no change)
1	0	0	$Q(t)$ (no change)
1	0	1	0
1	1	0	1
1	1	1	x

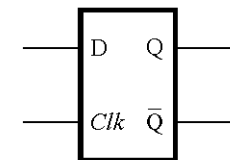
(b) Characteristic table



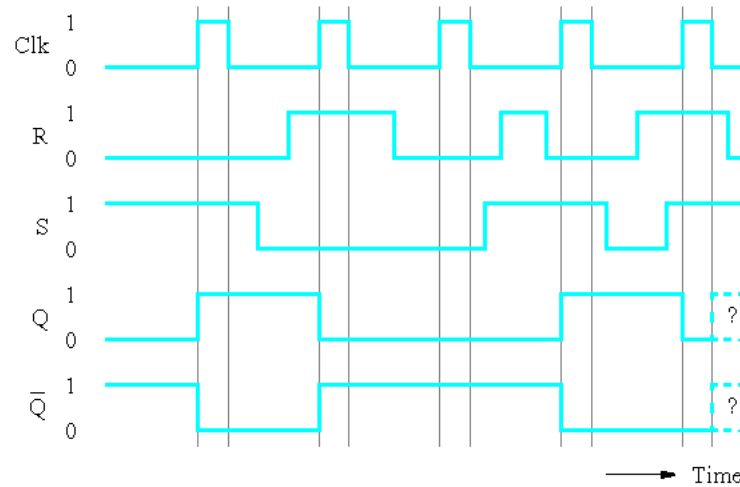
(a) Circuit

Clk	D	$Q(t+1)$
0	x	$Q(t)$
1	0	0
1	1	1

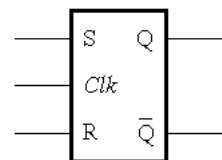
(b) Characteristic table



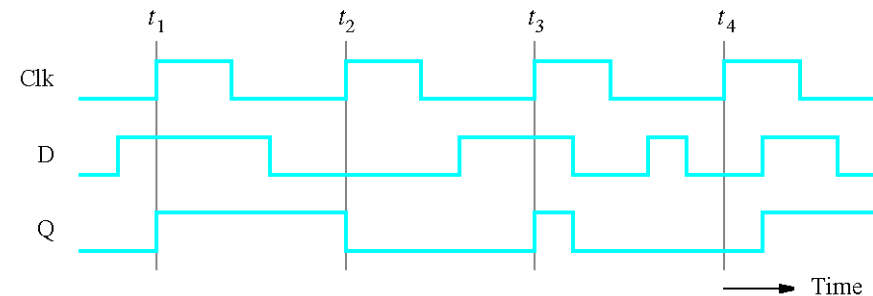
(c) Graphical symbol



(c) Timing diagram

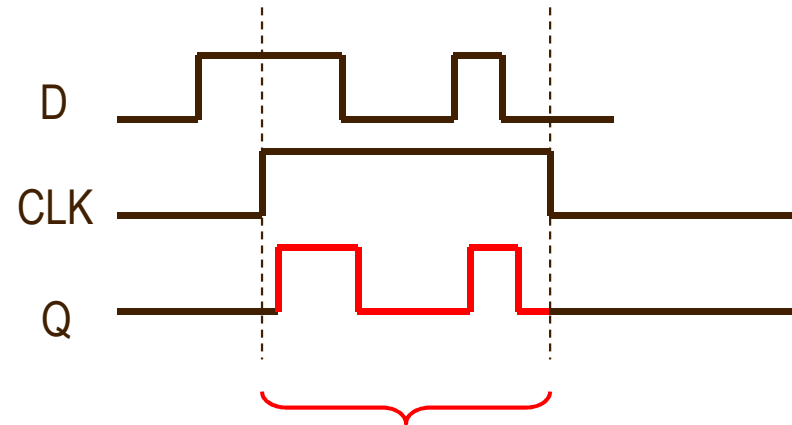
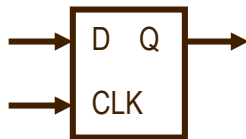


(d) Graphical symbol

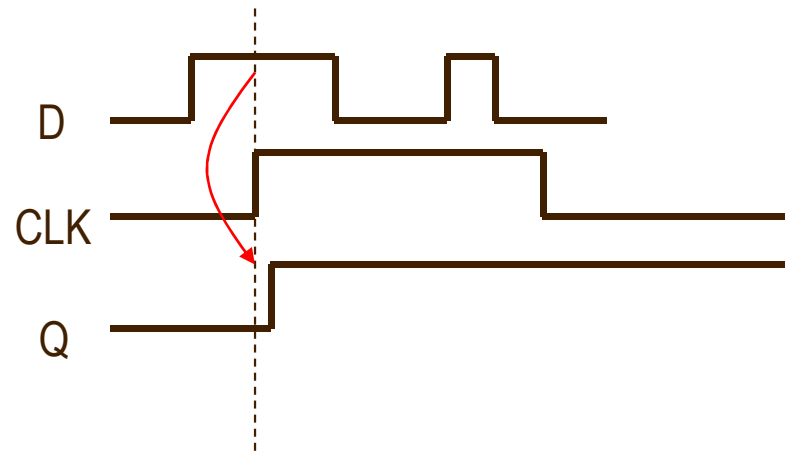
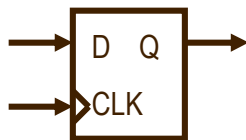


(d) Timing diagram

D Latch vs. D Flip-Flop

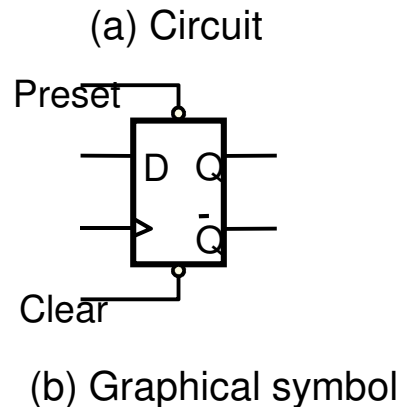


Latch transparent when clock is high



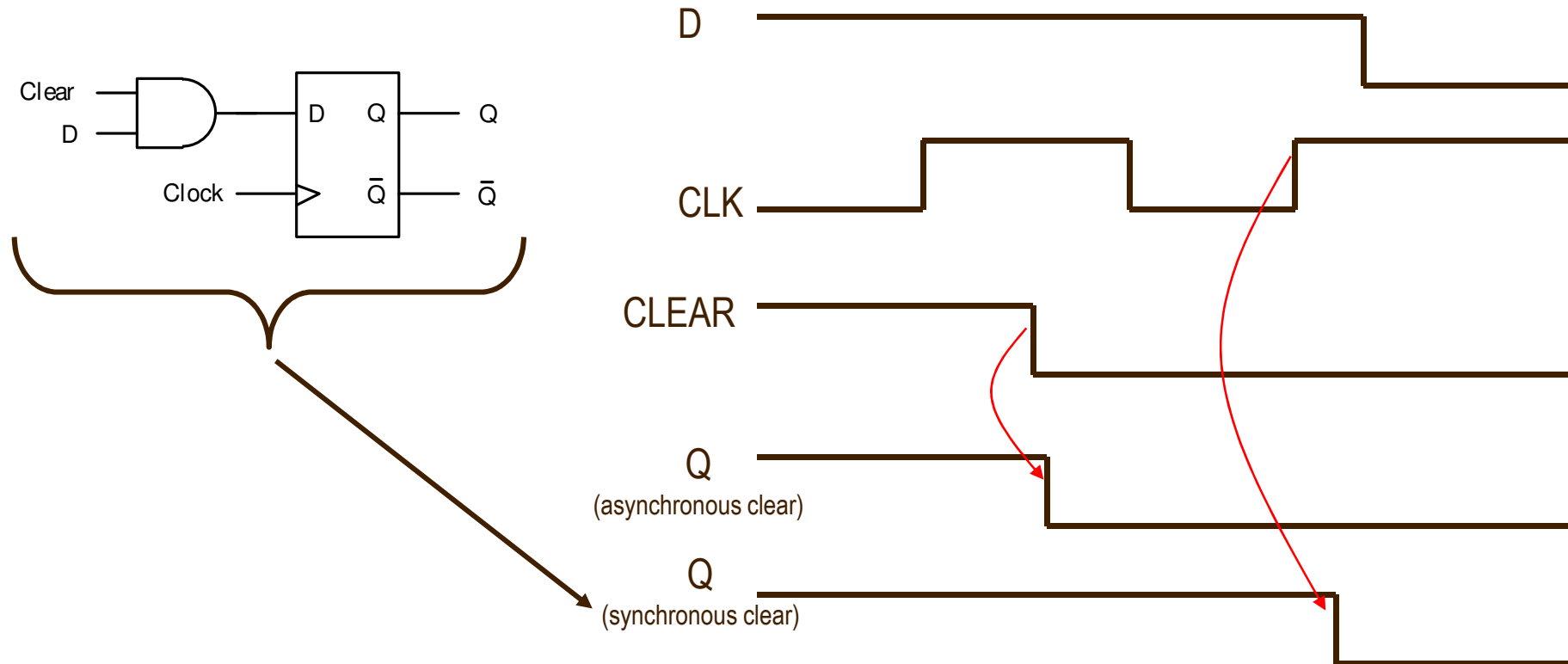
"Samples" D on rising edge of clock

D Flip-Flop with Asynchronous Preset and Clear

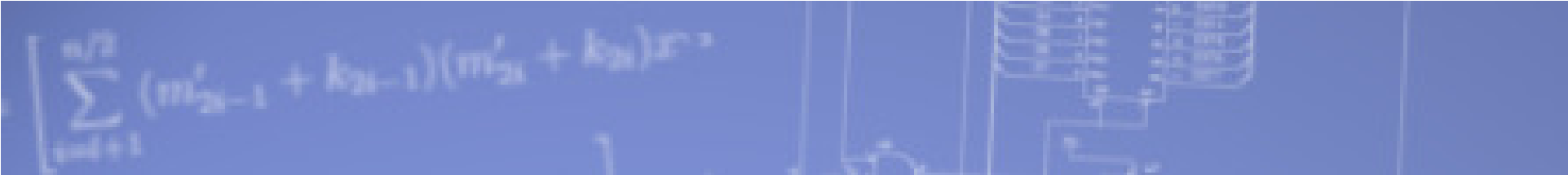


- Bubble on the symbol means “active-low”
 - When preset = 0, preset Q to 1
 - When preset = 1, do nothing
 - When clear = 0, clear Q to 0
 - When clear = 1, do nothing
- “Preset” and “Clear” also known as “Set” and “Reset” respectively
- In this circuit, preset and clear are asynchronous
 - Q changes immediately when preset or clear are active, regardless of clock

D Flip-Flop with Synchronous Clear

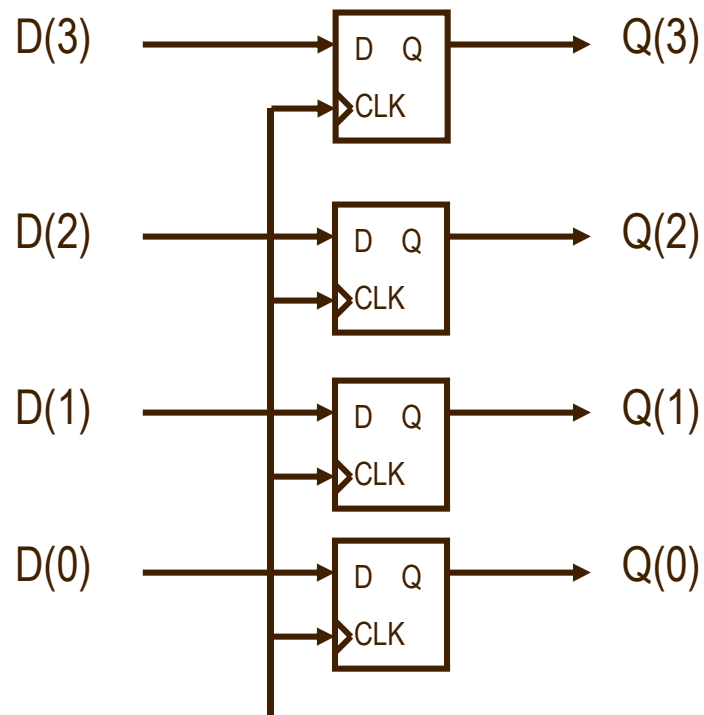


- Asynchronous active-low clear: Q immediately clears to 0
- Synchronous active-low clear: Q clears to 0 on rising-edge of clock



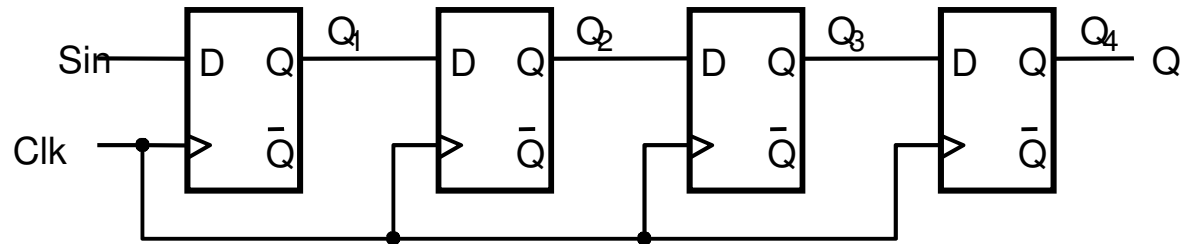
Sequential Logic Circuits

Register

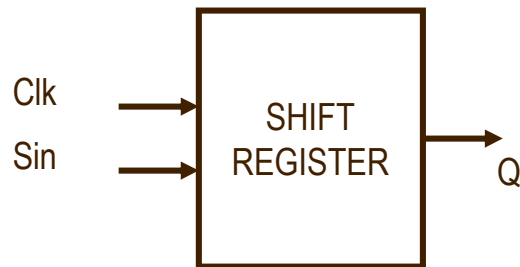


- In typical nomenclature, a register is a name for a collection of flip-flops used to hold a bus (i.e. `std_logic_vector`)

Shift Register



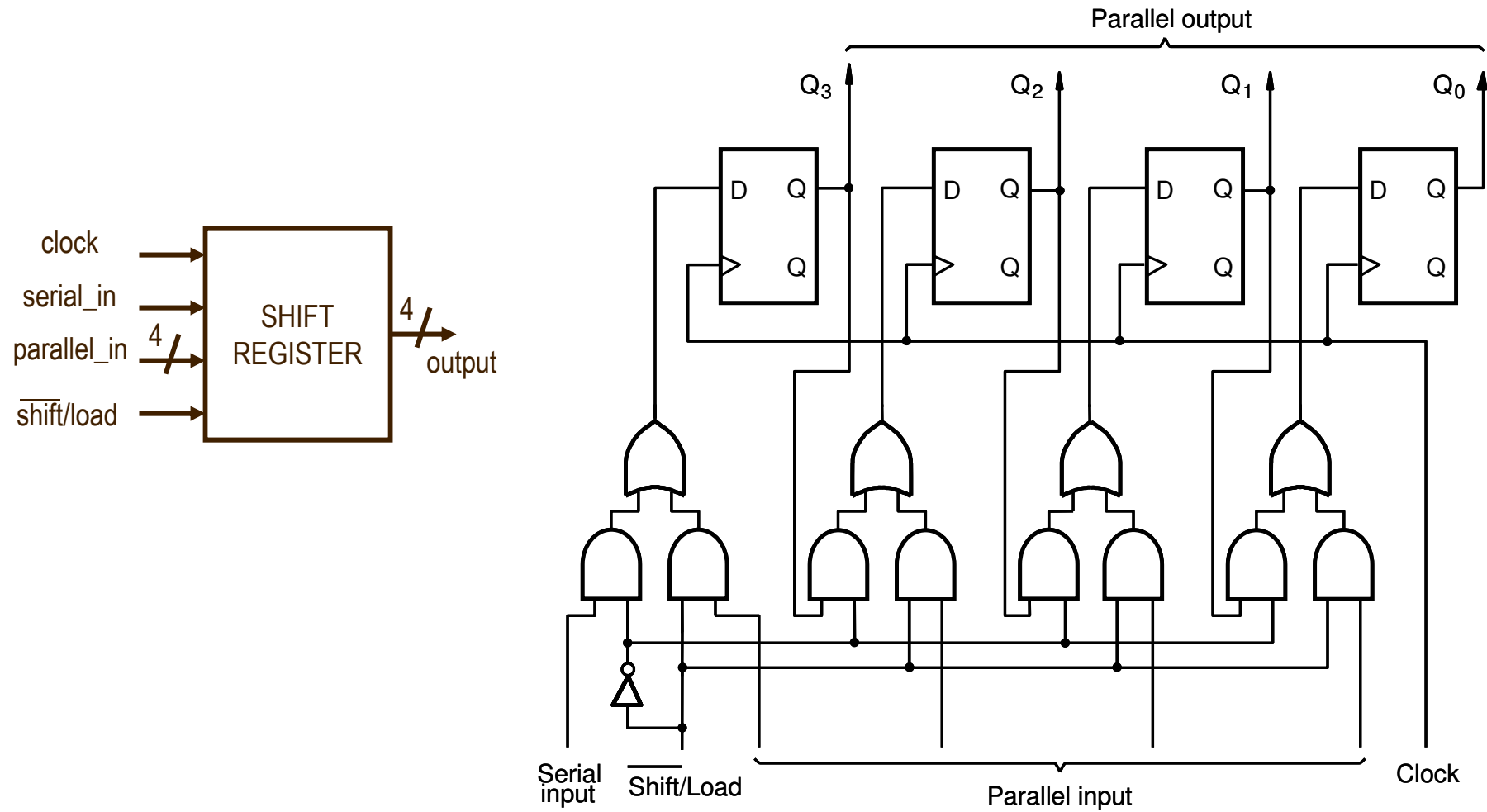
(a) Circuit



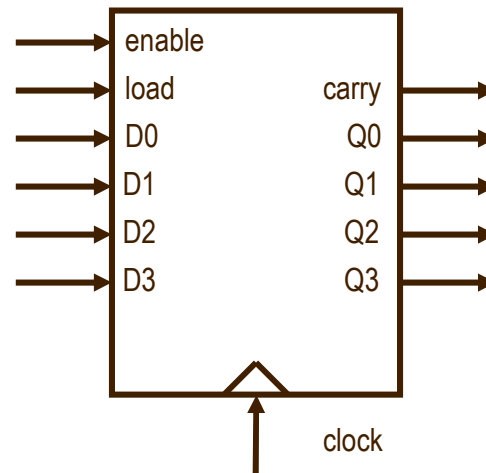
	Sin	Q_1	Q_2	Q_3	$Q_4 = Q$
t_0	1	0	0	0	0
t_1	0	1	0	0	0
t_2	1	0	1	0	0
t_3	1	1	0	1	0
t_4	1	1	1	0	1
t_5	0	1	1	1	0
t_6	0	0	1	1	1
t_7	0	0	0	1	1

(b) A sample sequence

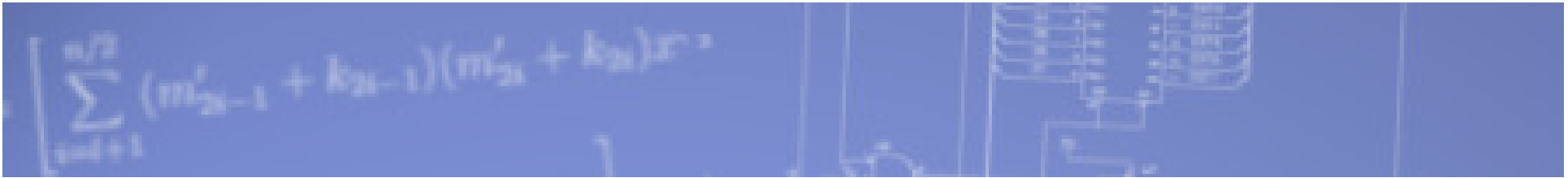
Parallel Access Shift Register



Synchronous Up Counter



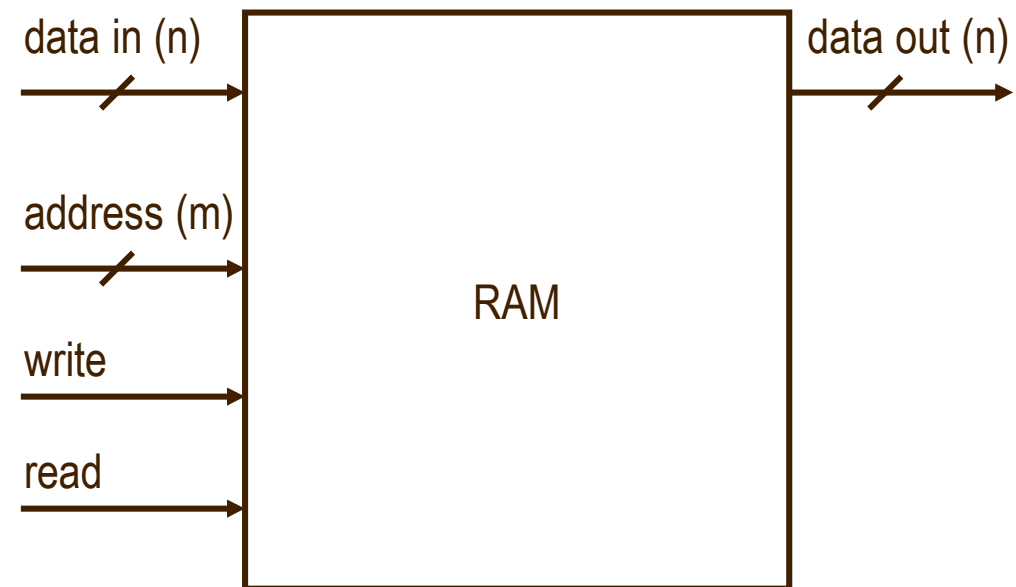
- Enable (synchronous): when high enables the counter, when low counter holds its value
- Load (synchronous) : when load = 1, load the desired value into the counter
- Output carry: indicates when the counter “rolls over”
- D3 down to D0, Q3 down to Q0 is how to interpret MSB to LSB



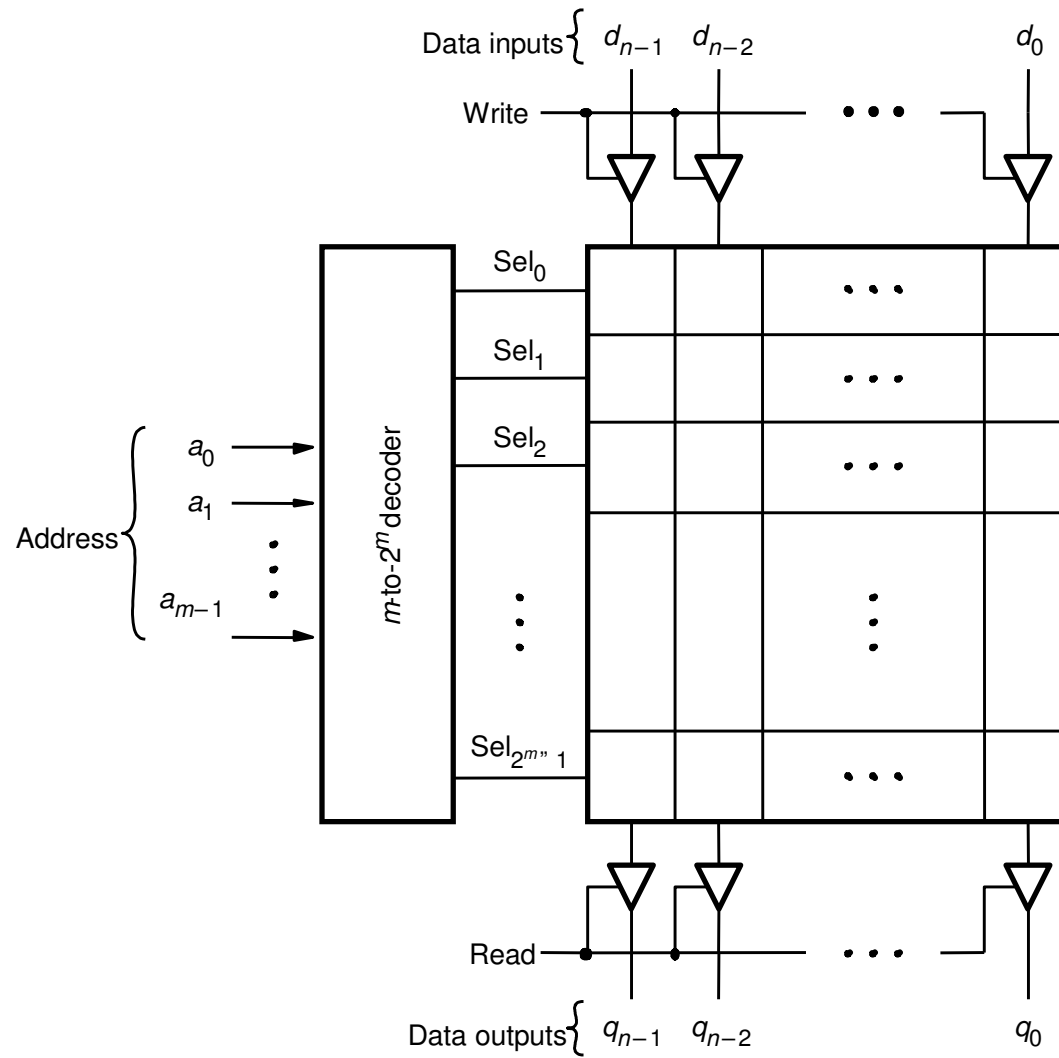
Memories

Random Access Memory (RAM)

- More efficient than registers for storing large amounts of data
- Can read and write to RAM
- Addressable memory
- Can be synchronous (with clock) or asynchronous (no clock)
- SRAM dimensions are:
 - (number of words) x (bits per word)
- SRAM
- Address is m bits, data is n bits
 - 2^m x n -bit RAM
- Example: address is 5 bits, data is 8 bits
 - 32 x 8-bit RAM
- Write
 - Data_in and address are stable
 - Assert write signal (then de-assert)
- Read
 - Address is stable
 - Assert read signal
 - Data_out is valid

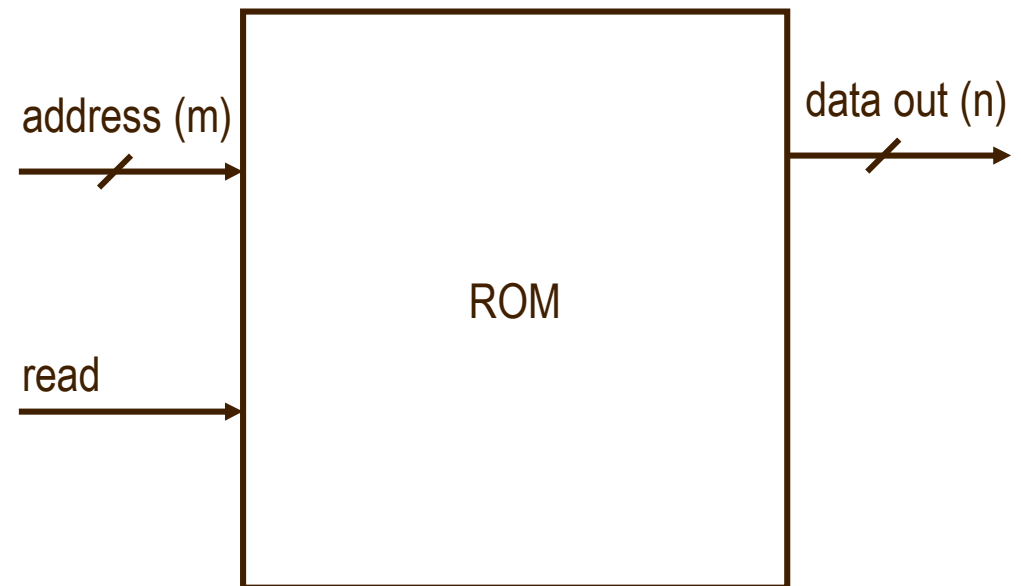


Random Access Memory (RAM)



Read Only Memory (ROM)

- Similar to RAM except read only
- Addressable memory
- Can be synchronous (with clock) or asynchronous (no clock)



Read-Only Memory (ROM)

